CAP250:DEEP LEARNING WITH GENERATIVE AI AND DATA SCIENCE

L:3 T:0 P:2 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: explain the fundamentals of AI, ML, Deep Learning, and the Data Science life cycle

CO2 :: apply deep learning models like feedforward, CNN, and RNN for real-world problems

CO3 :: analyze training challenges and evaluate model performance using suitable metrics

CO4 :: create generative models using frameworks like TensorFlow and PyTorch for practical applications

Unit I

Foundations of Deep Learning and Data Science : Introduction to AI, ML, Deep Learning, and Data Science, Overview of Data Science Life Cycle, Types of Learning- Supervised, Unsupervised, Reinforcement, Python for Data Science- Numpy, Pandas, Matplotlib, Data Preprocessing

Unit II

Feedforward neural networks : Pattern classification using perceptron, Multilayer feedforward neural networks, Backpropagation learning, Empirical risk minimization, Gradient-based Learning, Architecture Design, Computational Graphs and Regularization

Unit III

Advanced Deep Learning Architectures : Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), LSTM, GRU, Introduction to Transformers and Attention Mechanisms, Transfer Learning and Fine-Tuning Pre-trained Models

Unit IV

Deep and Convolutional Neural Networks (DNNs & CNNs) : Difficulty of training DNNs, Greedy layerwise training, Optimization for training DNNs, Regularization methods (dropout, batch normalization), Introduction, convolution, pooling, and Basic Convolution Function, Convolution Algorithm and Deep CNNs

Unit V

Sequence modelling : Recurrent neural networks (RNNs), Sequence modeling using RNNs, Long Short Term Memory (LSTM), Bidirectional LSTMs, Bidirectional RNNs, Gated RNN Architecture

Unit VI

Generative models : Boltzmann Machines, Restrictive Boltzmann Machines (RBMs), Stacking RBMs, Deep Boltzmann Machines, Applications in vision, speech and natural language processing, Deep Learning Frameworks- TensorFlow, PyTorch, Model Evaluation Metrics (Accuracy, Precision, Recall, F1, AUC)

List of Practicals / Experiments:

Practicals

- Load and explore a dataset using NumPy and Pandas.
- Perform data cleaning and preprocessing on a raw dataset.
- Implement a perceptron model for binary classification using Python.
- Train a basic CNN on the MNIST dataset using TensorFlow.
- Build an RNN or LSTM for text sentiment classification.
- Build a generative model using Restricted Boltzmann Machine (RBM).
- Implement and train a model using PyTorch and compare it with TensorFlow.

Text Books:

1. DEEP LEARNING by S. SRIDHAR, D. NARASHIMAN, PEARSON

References:

1. LEARNING DEEP LEARNING by EKMAN, PEARSON